

SOFTWARE REQUIREMENT SPECIFICATION (SRS)

Project Title

Secure Document Vault: A Role-Based Secure Document Management System

1. Introduction

1.1 Purpose

The Secure Document Vault is a web-based application designed to securely store, manage, and share digital documents within an organization. The system provides authentication, role-based access control, document upload/download functionality, audit logging, and secure document storage.

1.2 Scope

The system enables users to:

- Upload and manage documents securely.
- Access documents based on assigned roles.
- Maintain document confidentiality and integrity.
- Track user activities through audit logs.
- Prevent unauthorized access to sensitive documents.

1.3 Intended Audience

- System Administrators
 - Organization Employees
 - Project Evaluators
 - Developers and Maintenance Team
-

2. Overall Description

2.1 Product Perspective

Secure Document Vault is a centralized document repository developed as a web application using:

- Frontend: React.js
- Backend: Spring Boot
- Database: MySQL

2.2 Product Functions

- User Registration & Login
- JWT-Based Authentication
- Role-Based Access Control
- Document Upload
- Document Download
- Document Search
- Document Categorization
- Audit Logging
- User Management

2.3 User Classes

Administrator

- Manage users
- Assign roles
- View audit logs
- Manage documents

Employee/User

- Upload documents
- View permitted documents
- Download documents
- Search documents

2.4 Assumptions

- Users have internet access.
 - Documents are stored on the server.
 - MySQL database is available.
-

3. External Interface Requirements

3.1 User Interface

Login Page

- Username
- Password
- Login Button

Dashboard

- Upload Document
- View Documents
- Search Documents
- User Profile

Admin Panel

- Manage Users
- View Audit Logs
- Manage Access Rights

3.2 Hardware Interface

- Client PC/Laptop
- Server Machine
- Storage Device

3.3 Software Interface

Component	Technology
Frontend	React.js
Backend	Spring Boot
Database	MySQL
Security	JWT
API	REST API

4. System Features

4.1 User Authentication

- Secure login using username and password.

- JWT token generation.

4.2 Document Upload

- Upload PDF, DOCX, XLSX files.
- Metadata storage.

4.3 Document Download

- Authorized users can download files.

4.4 Role-Based Access Control

- Admin Role
- User Role

4.5 Document Search

- Search by name.
- Search by upload date.
- Search by category.

4.6 Audit Logging

Records:

- Login
 - Logout
 - Upload
 - Download
 - Delete
-

5. Non-Functional Requirements

Performance

- Response time less than 3 seconds.

Security

- Password Encryption.
- JWT Authentication.
- Role-Based Authorization.

Reliability

- 99% availability.

Scalability

- Support future cloud deployment.

Maintainability

- Modular architecture.
 - RESTful APIs.
-

6. Acceptance Criteria

The project shall be accepted if:

- ✓ Users can register and login successfully.
 - ✓ Documents can be uploaded and stored securely.
 - ✓ Documents can be downloaded only by authorized users.
 - ✓ User activities are logged.
 - ✓ Admin can manage users and permissions.
 - ✓ Search functionality works correctly.
-

7. Deliverables

1. Software Requirement Specification Document
2. Source Code
3. Database Schema
4. Test Cases
5. User Manual
6. Deployment Guide
7. Final Project Report
8. Presentation Slides